

Olympiad Test Paper — Social Science (Class 7) — Paper 5

Chapter 3: Climates of India

Paper 5 — (progressively harder)

Subject	Social Science (NCERT Class 7)
Chapter	3 — Climates of India
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: +4 for each correct answer · -1 for each wrong answer · 0 if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. The classic traps are treating one day's reading as climate, blaming equatorial heat on nearness to the Sun rather than ray-angle, forgetting that altitude and the sea both moderate temperature, reversing which monsoon is wet, mistaking the calm cyclone eye for safety, and assuming the windward and leeward sides of a range share the same rainfall. Do not write on this paper — mark answers on your answer sheet.

1. A town keeps thirty years of records. Year 18 was unusually rainy and warm; year 25 was unusually dry and cold. A researcher building the town's climate profile should treat these two odd years by:

- (A) discarding both years, because any year that differs from the others must be a measurement error
- (B) redefining the climate twice, once for the warm-wet pattern and once for the dry-cold pattern
- (C) including them in the multi-year average, because climate is precisely the long-run average that absorbs individual unusual years rather than being overturned by them
- (D) treating year 18 as the town's true climate because warm and wet is the most pleasant pattern

2. Two reports of the same desert town: Report 1 lists the actual rainfall of each of the last 40 years; Report 2 lists what fell on each day last week. Which report could, on its own, justify the sentence 'this town has an arid climate', and why?

- (A) Report 2, because the most recent week is the freshest and therefore the most reliable picture of climate
- (B) Either report equally, because rainfall figures of any length always describe climate
- (C) Report 1, because a 40-year span of yearly totals is long enough to reveal a stable average pattern, which is what climate describes
- (D) Neither report, because climate can never be inferred from rainfall data at all

3. A student writes: 'The weather of Chennai is hot and humid, and its climate this morning is sunny.' Identify the precise error.

- (A) There is no error, because weather and climate are interchangeable words for the same idea
- (B) The only error is that Chennai is not actually hot and humid, otherwise the sentence is well formed
- (C) The error is that a coastal city like Chennai cannot have weather, only climate
- (D) The two labels are swapped: a single morning's sunshine is weather, while the long-term hot-and-humid pattern is the climate

4. City G and City H lie at the same latitude; over thirty years each averaged 900 mm of rain and a mean temperature of 26 degrees. Yet one is described as having an even, humid climate and the other a savanna-like wet-and-dry climate. The most defensible reason these climates differ despite equal averages is that:

- (A) the two cities cannot really differ in climate if both averages match, so one description must be mistaken
- (B) the difference comes entirely from one city being slightly nearer the Sun than the other
- (C) thirty years is too short a record, so neither climate description can be trusted
- (D) averages alone hide how the rain and heat are spread through the year, and climate also depends on the seasonal distribution of those totals

5. A forecaster says 'tomorrow will be cooler than today.' A climatologist says 'this decade is warmer than the last.' Which statement is about weather, which about climate, and what single dimension separates them?

- (A) Both are about weather, because both mention temperature, and temperature is always a weather quantity
- (B) Both are about climate, because both compare one period with another
- (C) The forecaster speaks of weather and the climatologist of climate; the separating dimension is the length of time being described
- (D) The forecaster speaks of climate and the climatologist of weather, because predictions are always climate

6. Which of these would, by itself, FAIL to count as evidence about a place's climate?

- (A) the average July temperature computed over the past fifty years
- (B) the typical month-by-month rainfall pattern measured across decades

- (C) the usual spread between the hottest and coldest months observed over many years
- (D) a single afternoon thunderstorm recorded last Tuesday

7. At the equinox the Sun is overhead at the Equator. Comparing the heat delivered to one square metre of ground at the Equator with one square metre at 60 degrees north on that day, the equatorial square receives more energy chiefly because:

- (A) the equatorial square is millions of kilometres closer to the Sun than the polar square
- (B) the Sun emits stronger radiation toward the Equator than toward the poles
- (C) the polar square reflects every ray it receives, so it collects no energy at all
- (D) its near-vertical rays concentrate a beam onto a small patch, whereas the slanting polar rays spread the same beam over a larger patch

8. Madrid (about 40 degrees N) has hot summers and cold winters, while Quito (almost on the Equator) is mild and almost unchanging all year. The deepest reason for Quito's tiny seasonal swing is that:

- (A) near the Equator the Sun stays high in the sky every month, so the ray-angle and the heating it delivers barely change from season to season
- (B) Quito is at a lower altitude than Madrid, and low places always have steady temperatures
- (C) Quito is closer to the Sun in every season than Madrid is
- (D) the Equator has no atmosphere, so its temperature cannot vary

9. A pupil reasons: 'Summer is hot because in summer the Earth is closest to the Sun in its orbit.' For India, the strongest objection is that:

- (A) the objection fails, because orbital distance really is what causes the seasons
- (B) the Earth's orbit is a perfect circle, so its distance from the Sun never changes at all
- (C) seasons are caused by the monsoon winds reversing, with the Sun playing no part
- (D) if mere orbital distance set the seasons, the whole planet would be hot at once, yet India's summer coincides with winter in the southern hemisphere, so seasonal heat must come from the tilt and ray-angle instead

10. Place P is at 10 degrees N, Place Q at 50 degrees N, both at sea level on the coast. Over a year, which prediction is best supported by the latitude factor alone?

- (A) Q will be warmer overall, because higher latitudes catch the Sun for more total hours every year
- (B) both will have identical temperatures, since both are coastal and at sea level
- (C) P will be warmer overall and have a smaller seasonal swing, because its higher Sun keeps heating steady and strong all year
- (D) P will have the larger seasonal swing, because places near the Equator change most between seasons

11. Two students explain why the tropics stay warm in every season. Student 1: 'because the tropics are permanently tilted toward the Sun.' Student 2: 'because the Sun stays high in the tropical sky all year, so rays strike steeply each season.' Whose reasoning is sound, and why?

- (A) Student 2, because the Earth's tilt is fixed in space and the tropics are not permanently aimed at the Sun; what keeps them warm is the steep ray-angle maintained throughout the year
- (B) Student 1, because the tropics genuinely lean toward the Sun in every month
- (C) Both, because the two statements say exactly the same thing in different words
- (D) Neither, because tropical warmth is caused only by nearness to the sea

12. Why does a slanting ray near the pole deliver less warmth than a vertical ray at the Equator, for two reasons working together?

- (A) the slanting ray travels faster and therefore loses heat, and it also misses the ground entirely
- (B) the slanting ray is colder light to begin with, and the polar ground refuses to absorb it
- (C) the slanting ray is bent upward by the cold air and the Sun is dimmer over the poles
- (D) the slanting beam is spread over a larger ground area AND it crosses a longer path of atmosphere that scatters and absorbs more of its energy

13. Town X is at 28 degrees N and 2,000 m altitude; Town Y is at 28 degrees N and 100 m altitude, both far inland. A pupil insists X must be warmer 'because higher means nearer the Sun.' The correct prediction and its reason are:

- (A) X is cooler, because temperature falls with height as the air thins and lies farther from the ground that warms it, and the few kilometres of extra height toward the Sun are negligible
- (B) X is warmer, because the pupil is right that altitude brings a place measurably closer to the Sun
- (C) X and Y are equal, because their shared latitude fixes their temperature
- (D) X is cooler, but only because higher places drift closer to the poles

14. On one tall tropical mountain the base is steaming jungle and the summit carries permanent snow, all at the same latitude within a few horizontal kilometres. Which explanation is correct?

- (A) altitude alone explains it: temperature falls steadily with height, so the same latitude can hold jungle at the base and snow at the top
- (B) latitude explains it, because latitude changes sharply over those few kilometres of climb

(C) distance from the sea explains it, because the summit is much farther from the ocean than the base

(D) nearness to the Sun explains it, because the snowy summit is closest to the Sun and so coldest

15. Air temperature drops about 6.5 degrees for every 1,000 m of ascent. A hill station sits 1,800 m above a plain whose mean temperature is 32 degrees. The hill station's expected mean temperature, and the reason it is cooler, are about:

(A) around 44 degrees, warmer because rising 1,800 m brings the station noticeably closer to the Sun

(B) around 20 degrees, cooler because the air thins and lies farther from the ground that heats it as height increases

(C) around 32 degrees, unchanged because altitude does not affect temperature at the same latitude

(D) around 12 degrees, cooler because higher places are nearer the cold poles

16. Coastal City J and interior City K share latitude and altitude. Over the year City J ranges from 22 to 31 degrees; City K ranges from 8 to 42 degrees. Which single factor best accounts for K's far wider range, and through what mechanism?

(A) latitude, since K must secretly lie much farther from the Equator despite the figures given

(B) continentality: far from the slow-changing sea, K's land heats strongly in summer and loses its heat strongly in winter, pushing both extremes apart

(C) altitude, since interiors are always perched much higher than coasts

(D) the sea actively pumping heat into J in summer and cold into J in winter

17. Why does continentality push a single interior place toward BOTH hotter summers AND colder winters, rather than just one extreme?

(A) the interior physically moves nearer the Sun in summer and farther in winter

(B) the distant sea heats the interior in summer and chills it in winter from afar

(C) the interior changes its latitude between the two seasons

(D) without the slow-changing sea nearby, the land heats up strongly under the summer Sun and loses that heat strongly in winter, so both ends of the range are pushed outward

18. Two coastal cities lie at the same latitude and altitude, yet one shows a slightly wider yearly temperature range than the other. The most reasonable extra factor to investigate, staying within the chapter's ideas, is:

(A) which city happens to be marginally closer to the Sun

(B) how exposed each city is to onshore sea winds, since a city sheltered from the moderating sea air will swing a little more

(C) which city has the larger human population this year

(D) which city the Moon passes over more frequently

19. Mumbai (coastal) and Nagpur (deep inland) lie near the same latitude. Which statement correctly pairs each city's summer behaviour with the right cause?

- (A) Mumbai has fiercer summers because the sea reflects extra heat onto it, while Nagpur stays mild inland
- (B) both have identical summers, because the shared latitude alone fixes the heat they receive
- (C) Mumbai has milder summers because the slow-warming sea holds its temperature down, while Nagpur has fiercer summers because inland land heats rapidly with no sea to steady it
- (D) Nagpur is milder because being inland makes it nearer the Equator

20. A place with an unusually SMALL gap between its warmest and coolest months, all else typical, most likely owes that smallness to:

- (A) sitting deep in a dry continental interior far from any sea
- (B) lying close to the moderating sea, whose slow heating and cooling evens out the temperature across the year
- (C) perching on a very high, exposed plateau remote from any coast
- (D) lying at a very high latitude where the Sun's angle swings strongly between seasons

21. The hot, dry 'loo' of north-west India in summer and the chilling winds that bring winter cold waves to the plains differ greatly, yet they teach the SAME general lesson about winds, which is that:

- (A) a wind carries the temperature and moisture qualities of whatever land or sea it has just crossed
- (B) winds change a region's latitude as they blow across it
- (C) winds raise or lower the altitude of the land they pass over
- (D) winds physically push a region nearer to or farther from the sea

22. Wind A has just crossed a warm sea before reaching a coast; Wind B has just crossed a hot, sandy desert. Reaching the same town on the same hot day, the most reliable contrast is that Wind A is:

- (A) moister than Wind B, because it picked up water vapour while crossing the warm sea, whereas Wind B stayed dry over the desert
- (B) drier than Wind B, because seas remove water vapour from any air crossing them
- (C) colder than Wind B in every case, because seawater is always icy
- (D) identical to Wind B, because the surface a wind crosses leaves no mark on it

23. A cold wave reaches the northern plains when winter winds spill over the Himalayas, while at the same season a wind crossing the Bay of Bengal toward the Tamil Nadu coast is mild and damp. The difference between these two winds is best explained by:

- (A) the Tamil Nadu wind being at a far lower latitude than the Himalayan wind never reaches
- (B) the Himalayan wind being heavier and therefore always colder than any sea wind

- (C) the two winds being the same air that simply changes its mind about temperature
- (D) the source surfaces they crossed: cold snowy mountains for one, a warm sea for the other, so each carries the quality of its path

24. A flat, open plain and a valley shielded behind a high ridge sit at the same latitude. In summer the open plain bakes under a dry continental wind while the sheltered valley is spared the worst of it. This contrast chiefly reveals the climate role of:

- (A) latitude, since the two places must lie at different distances from the Equator
- (B) distance from the sea, since one place must be far nearer the ocean than the other
- (C) the position of the Sun, regardless of the shape of the land
- (D) relief (topography), since open flat ground exposes a place while a mountain barrier shelters the land behind it

25. The 'loo' would be cool and moist instead of hot and dry only if it were to arrive after crossing:

- (A) an even larger and hotter desert than it actually crosses
- (B) the snow-free summer plains where it gains still more heat
- (C) a wide stretch of cool sea, from which it would pick up moisture and lose its searing heat
- (D) no land or sea at all, travelling instead through empty space

26. Moist monsoon air is forced up the windward slope of the Western Ghats. Order the physical changes that wring rain from it:

- (A) the air rises and is compressed, warming so it holds more vapour and stays rainless
- (B) the air rises, speeds up and dries out instantly without any cooling
- (C) the air rises and expands, cooling as it does; cooler air holds less water vapour, so the vapour condenses into cloud and falls as rain
- (D) the air rises, then sinks back down the same windward slope and rains there a second time

27. The Deccan Plateau lies just east of the Western Ghats and is comparatively dry. Reordered as a clean cause-and-effect chain, the rain-shadow story runs:

- (A) the Deccan lies nearer the Equator, heats up, and repels the rain clouds before they arrive
- (B) moist winds meet the Ghats, are lifted and rain on the windward (western) slopes, then cross the crest as dry, descending air, leaving the leeward Deccan in a rain shadow
- (C) the west coast stands higher than the Ghats, so the rain collects on the coast first
- (D) the Deccan has no rivers, so no rain can form in its sky

28. Ladakh, beyond the high Himalayas, is described as a 'cold desert.' Which pairing correctly explains BOTH halves of that label?

- (A) it is dry because the mountains block the rain-bearing winds, and cold because its very great altitude keeps temperatures low
- (B) it is dry because the nearby sea pulls the moisture away, and cold because it lies right at the North Pole
- (C) it is hot and dry like the Thar, with 'cold' being only a nickname
- (D) it is cold because of low latitude and dry because the rivers there evaporate

29. Cherrapunji and Mawsynram receive among the heaviest rainfall on Earth. The chain that best explains this is that the monsoon winds there are:

- (A) dry winds off the Thar that somehow carry the desert's rain to the hills
- (B) winds descending the leeward slopes and warming as they fall, which makes them rain
- (C) moisture-laden after crossing the Bay of Bengal and then funnelled and forced steeply up against the hills, where they cool and dump torrential rain
- (D) cold winter winds off the Himalayas dropping their snow as rain in summer

30. On a July rainfall map the windward Konkan coast is soaked, the Deccan behind it is dry, the north-eastern hills are drenched, and the valley below them is less wet. The single idea that ties ALL four observations together is:

- (A) latitude alone, since rainfall simply increases as one nears the Equator
- (B) relief lifting moist monsoon winds, giving heavy rain where the air is forced up and far less in the rain shadow or lower ground behind
- (C) altitude alone, since higher land is always wetter than lower land everywhere
- (D) distance from the sea alone, since the farther inland one goes the wetter it always becomes

31. If the Western Ghats were entirely flattened, the most likely change to rainfall would be that:

- (A) the west coast would rain even more heavily than before, with the barrier gone
- (B) the whole peninsula would turn into frozen tundra
- (C) rainfall everywhere would remain exactly as it is now
- (D) the west coast would lose much of its heavy rain, since nothing would lift and wring out the moist winds, while more moisture might pass on to the now less-shadowed interior

32. A hiker crosses a coastal ridge: the seaward climb is misty and dripping, but within a few kilometres the far valley is sunny and parched. The best explanation for this rapid switch is:

- (A) the rain-shadow effect, as moist air rains itself out on the windward seaward slope and descends dry and warming onto the leeward valley
- (B) a sudden large jump in latitude over those few kilometres

- (C) the far valley lying much closer to the Equator than the seaward slope
- (D) the two sides being made of different soils that decide the rainfall

33. The whole Indian monsoon ultimately rests on one physical fact about land and water. Removing which fact would make the monsoon impossible?

- (A) that land heats up and cools down faster than the ocean, so seasonal pressure differences build between them
- (B) that the Moon's gravity drags monsoon clouds toward India each summer
- (C) that rivers push warm air upward and pull sea winds inland
- (D) that the Equator stays at one fixed temperature all year

34. In summer the interior of Asia heats intensely. The pressure that forms there and the wind it draws are correctly described as:

- (A) high pressure over the hot land, which pushes dry winds out to sea
- (B) low pressure over the hot land, into which dry winds blow outward to the sea
- (C) low pressure over the hot land, into which moisture-laden sea winds blow from the south-west
- (D) high pressure over the hot land, which draws moist sea air inland

35. Why does the south-west monsoon bring India's main rains while the north-east monsoon mostly does not?

- (A) the south-west winds come from the dry Thar while the north-east winds come off the sea
- (B) the south-west winds cross warm seas and arrive moisture-laden, while the north-east winds blow off the dry land toward the sea
- (C) the south-west winds are colder, and cold winds always carry more rain
- (D) the south-west winds fall straight down from the upper atmosphere carrying rain

36. Most of the north-east winter monsoon is dry, yet the south-eastern coast of Tamil Nadu gets notable rain at that time. The reason is that, before striking that coast, the winter winds have:

- (A) crossed the Bay of Bengal and picked up moisture there
- (B) crossed the Thar Desert and gathered its sand-laden moisture
- (C) blown straight down the Himalayas carrying snow-melt as rain
- (D) come fresh off the Arabian Sea from the west

37. The deepest reason the monsoon winds reverse between summer and winter is that:

- (A) the Earth tilts its axis the opposite way twice each year
- (B) the land-sea pressure pattern flips, since the land is a low-pressure zone in hot summer and a high-pressure zone in cold winter
- (C) the Western Ghats physically swing around with the seasons
- (D) the Moon switches the direction of its orbit each season

38. Place the phases of the Indian monsoon year in order, starting from the dry winter: hot dry summer, rainy south-west monsoon, retreating monsoon, dry winter. The correct cycle is:

- (A) dry winter, then rainy south-west monsoon, then hot dry summer, then retreating monsoon
- (B) dry winter, then retreating monsoon, then rainy south-west monsoon, then hot dry summer
- (C) dry winter, then hot dry summer, then retreating monsoon, then rainy south-west monsoon
- (D) dry winter, then hot dry summer, then rainy south-west monsoon, then retreating monsoon

39. The summer monsoon is described as 'a sea breeze the size of a country.' The comparison holds because both a daytime sea breeze and the summer monsoon are driven by:

- (A) the Moon's gravitational pull on ocean water
- (B) warm land creating low pressure that pulls in cooler, moister air from over the sea
- (C) rivers warming the air directly above their channels
- (D) snow melting on high mountains and releasing winds

40. The monsoon usually reaches Kerala in early June and then advances. The correct description of its spread is that it moves:

- (A) from the southern tip northward, covering the whole country by about mid-July
- (B) from the far north southward, reaching Kerala last in September
- (C) from the east coast outward into the surrounding seas
- (D) everywhere across the country at the same instant in early June

41. The 'burst' of the monsoon feels dramatic and abrupt because:

- (A) the Sun suddenly stops shining over India for the season
- (B) the rivers suddenly rise and spill their water into the air as rain
- (C) the rain-bearing south-west winds arrive in force and unleash steady heavy rain after weeks of mounting pre-monsoon heat and dryness
- (D) the dry north-east winter monsoon arrives all at once from the north-east

42. The 'retreating monsoon' season is best defined as the period when:

- (A) the south-west monsoon first bursts over Kerala
- (B) the south-west monsoon weakens and withdraws, bridging the rains and the dry winter
- (C) the summer heat first builds up before any rain has fallen
- (D) the rivers retreat back up into the Himalayas

43. India is often said to have a 'fifth season' beyond the usual four. The reason the rainy season is singled out is that:

- (A) India copied a five-season calendar directly from Europe
- (B) India has no winter at all, leaving an empty slot for a fifth season
- (C) the monsoon rains dominate Indian life so strongly that they form a distinct season of their own
- (D) every Indian festival demands its own separate season

44. Why do the six traditional ritis divide the Indian year more finely than the four-season scheme?

- (A) they were invented only recently for the tourist trade
- (B) they ignore the monsoon entirely and follow the Sun alone
- (C) they track the monsoon-ruled year more closely, recognising transitions that the four-fold scheme lumps together
- (D) they are simply the four European seasons given new names

45. Festivals such as Bihu, Pongal, Onam and Baisakhi are tied to the seasons and the farming cycle. What does this connection most reveal about climate?

- (A) climate, through the monsoon and seasons, shapes farming and therefore the timing and meaning of cultural life
- (B) festivals control the climate by deciding when the rains will arrive
- (C) climate has no effect whatever on human culture in India
- (D) festivals depend only on each state's latitude and nothing else

46. Why do the same seasons return year after year in India?

- (A) the Sun grows hotter every summer and cooler every winter
- (B) the Earth revolves around the Sun on its tilted axis, so the same pattern of heating repeats every year
- (C) the monsoon winds choose to reverse on fixed festival dates
- (D) rivers flood at the same time each year and thereby create the seasons

47. Srinagar (far north, in the mountains) is far colder than Kanniyakumari (far south, by the sea). Which TWO factors combine most powerfully to make Srinagar so cold?

- (A) its low latitude and its nearness to the sea
- (B) its high latitude, which gives a weaker, more slanting Sun, together with its great altitude, both pushing temperatures down
- (C) its position right on the Equator with the Sun constantly overhead
- (D) the moist warm sea winds that blow steadily over it all year

48. Match each region to the factor that most powerfully shapes its climate: (i) Ladakh's cold desert, (ii) Mumbai's mild range, (iii) Cherrapunji's deluge, (iv) Kanniyakumari's year-round warmth. The best matching is:

- (A) (i) high altitude plus rain shadow, (ii) nearness to the moderating sea, (iii) relief lifting moist winds, (iv) low near-equatorial latitude
- (B) (i) low latitude, (ii) high altitude, (iii) distance from the sea, (iv) relief

- (C) (i) nearness to the sea, (ii) latitude, (iii) altitude alone, (iv) winds only
- (D) (i) winds only, (ii) rain shadow, (iii) low latitude, (iv) altitude

49. 'Monsoon failure' means the seasonal rains are weak or insufficient. The most likely chain of consequences for India is:

- (A) rivers overflow, widespread floods spread, and harvests boom
- (B) crops wither, water grows scarce, harvests shrink, and food prices rise
- (C) winters turn much warmer, glaciers grow, and water becomes plentiful
- (D) the six ritus vanish and the seasonal calendar collapses

50. Why does a weak monsoon hit India's economy especially hard?

- (A) festivals are cancelled by government order during dry years
- (B) the sea stops moderating the coastal cities when the rains fail
- (C) the Himalayas block trade routes whenever it does not rain
- (D) a large share of farming depends on monsoon rain, so poor rains cut harvests, reduce rural incomes, and raise food prices

51. Trace the chain from a distant Pacific cause to a local Indian effect: 'warm Pacific water shifts eastward, rearranging tropical winds, leading to ____.' The most likely next link is:

- (A) record floods across India that very year
- (B) the Himalayas growing several metres taller
- (C) a weakened monsoon, so India receives less rain and crops and water supplies are stressed
- (D) the sea ceasing to moderate India's coastline

52. Every few years the monsoon is disrupted by a periodic warming of the central and eastern Pacific Ocean that rearranges tropical winds. This recurring event is known as:

- (A) El Nino
- (B) the loo
- (C) a rain shadow
- (D) the burst of the monsoon

53. A cyclone forms over warm seas when the air pressure at its centre becomes:

- (A) much higher than its surroundings, pushing air away from the centre
- (B) much lower than its surroundings, so air and moisture rush in and spiral upward
- (C) exactly equal to its surroundings everywhere
- (D) colder than the land in the middle of winter

54. During a cyclone the wind suddenly drops and the sky clears overhead. The safest interpretation is that:

- (A) the storm has fully ended and it is now safe to go outside
- (B) the cyclone has turned into a harmless breeze
- (C) the calm eye is the most destructive part and one must flee through it
- (D) the calm eye is passing over, and fierce winds will soon return from the opposite direction once it moves on

55. Where in a cyclone are the most violent winds found, and why are people there in the greatest danger?

- (A) in the calm, clear eye at the very centre, where the storm is strongest
- (B) in the eyewall, the dense ring of cloud surrounding the calm centre, where the strongest winds whirl
- (C) hundreds of kilometres out beyond the edge of the storm
- (D) only after the storm has completely passed and the air is still

56. Why does the calm 'eye' of a cyclone trick people into venturing outside?

- (A) the eye is in fact the part with the strongest winds, which merely feel calm
- (B) once the eye arrives the cyclone can never strengthen again
- (C) the eye marks the point where the cyclone has left the sea for good
- (D) the eye brings a sudden lull with clear sky, but the fierce eyewall winds return once it moves on

57. Why does the NATURAL greenhouse effect actually make life on Earth possible?

- (A) they block all of the Sun's rays, keeping Earth cool
- (B) they cool the planet to below freezing, which life prefers
- (C) greenhouse gases trap part of the Sun's heat, keeping the planet warm enough for living things
- (D) they directly create the monsoon winds that water the land

58. Place these links in correct order for the chain 'a factory burns coal, so a distant farmer's monsoon crop fails':

- (A) coal burnt, sea cools, monsoon strengthens, crop thrives
- (B) coal burnt, rivers freeze, no irrigation, crop fails
- (C) coal burnt, forests grow, more rain, crop floods
- (D) coal burnt, carbon dioxide released, extra heat trapped, climate shifts, monsoon disrupted, crop fails

59. Which single action is the clearest example of climate mitigation rather than resilience?

- (A) replacing coal power with solar and wind energy to cut greenhouse-gas emissions
- (B) building higher sea walls to protect a coastal town from storm surges
- (C) stockpiling food and water so a village can survive a drought
- (D) moving homes away from a flood-prone riverbank

60. Since the nineteenth century, the main driver of accelerating climate change has been:

- (A) natural processes acting entirely on their own
- (B) the Earth slowly drifting farther from the Sun
- (C) human activities, especially the burning of fossil fuels
- (D) volcanic eruptions and nothing else

Solutions — Social Science (Class 7), Chapter 3: Climates of India — Paper 5

Paper 5 — (progressively harder)

Marking: +4 correct, –1 wrong, 0 unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	C	11	A	21	A	31	D	41	C	51	C
2	C	12	D	22	A	32	A	42	B	52	A
3	D	13	A	23	D	33	A	43	C	53	B
4	D	14	A	24	D	34	C	44	C	54	D
5	C	15	B	25	C	35	B	45	A	55	B
6	D	16	B	26	C	36	A	46	B	56	D
7	D	17	D	27	B	37	B	47	B	57	C
8	A	18	B	28	A	38	D	48	A	58	D
9	D	19	C	29	C	39	B	49	B	59	A
10	C	20	B	30	B	40	A	50	D	60	C

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (C) Climate is the average condition over a long span, typically decades. The whole point of averaging across many years is that no single year, however unusual, defines the climate; the odd years are data points that get smoothed into the long-run pattern, so they should be kept in the average. Discarding them as 'errors' is wrong, because a genuinely recorded extreme year is real weather data, not a mistake. Splitting the record into two separate climates confuses individual years with climate, which is a single long-term summary. Picking one pleasant year as 'the climate' abandons averaging entirely and substitutes preference for evidence.

2. (C) Climate is the long-term average, so the record that spans decades (40 years of totals) is exactly the kind of evidence that can establish a climate label like 'arid'. The week-long record covers only current weather, so freshness does not make it a climate record. The claim that rainfall data 'of any length' shows climate ignores the time-scale requirement; a single week cannot. The claim that rainfall never reveals climate is false, since long-run rainfall averages are a core part of how climate is defined.

3. (D) Weather is the condition at a moment or over a short span; climate is the long-term average. 'This morning... sunny' is a momentary condition, so it is weather, and 'hot and humid' as a year-round descriptor is the climate, so the student has attached each label to the wrong idea. The words are not interchangeable; their defining difference is the time-scale. Whether Chennai is in fact hot and humid is beside the point, since the structural mistake is the swapped labels. Every place has both weather and climate, so coastal location does not remove either.

4. (D) Two places can share identical annual averages yet differ in climate because climate includes not just totals but how temperature and rainfall are distributed across the seasons; one may spread its 900 mm evenly while the other crams it into a wet season and bakes the rest of the year. So equal averages do not force identical climates. Distance from the Sun barely varies between two cities at the same latitude and is never the controlling factor. Thirty years is a perfectly adequate climate record; the issue is distribution within the year, not the length of the record.

5. (C) A day-to-day forecast is weather; a comparison of one decade with another is climate, and the one thing that distinguishes them is the time-scale, not the quantity measured. Temperature appears in both kinds of statement, so mentioning it does not make a statement weather. Comparing two periods does not by itself make a statement climate, since a forecaster also compares today with tomorrow. Predicting the future is not the same as describing climate, so calling the forecast 'climate' inverts the definitions.

6. (D) Climate is built from long-term averages and typical patterns, so evidence about climate must span many years. A single afternoon storm is one short event, which is weather, not climate. The fifty-year average July temperature, the decades-long month-by-month rainfall pattern, and the usual hot-to-cold spread observed over many years are all multi-year summaries, and each is exactly the kind of evidence that does describe climate.

7. (D) A near-vertical beam at the Equator lands on a small patch, so the energy per square metre is high; the same beam striking at a slant near 60 degrees is smeared across a larger patch, so each square metre gets less. The difference is the angle, not distance: the few-thousand-kilometre gap between the patches is negligible against the Sun's 150-million-kilometre distance. The Sun does not aim stronger radiation at the Equator; it radiates in all directions. The poles do not reflect every ray, and even bare polar ground would still receive less because of the slant.

8. (A) Because the Earth's tilt swings the Sun's overhead point only within the tropics, an equatorial place keeps the Sun high all year, so the angle and intensity of heating hardly change between months; that steadiness gives Quito a tiny seasonal range. Altitude does not produce steadiness, and Quito in fact sits high in the Andes, so the low-altitude claim is factually wrong as well as irrelevant. Distance from the Sun barely differs between places on Earth, so it cannot explain the contrast. The Equator certainly has an atmosphere, so that option is simply false.

9. (D) If closeness to the Sun caused summer, both hemispheres would be hot together, but India's summer is the southern hemisphere's winter; the two hemispheres have opposite seasons because each is tilted toward or away from the Sun in turn, which changes the ray-angle. So seasons come from tilt, not orbital distance. Endorsing the pupil is therefore wrong. The orbit is slightly elliptical rather than a perfect circle, but the key flaw is the simultaneity argument, not the orbit's shape. The monsoon reversal is itself a consequence of seasonal heating, not the cause of the seasons.

10. (C) The lower-latitude place (P, 10 degrees N) keeps the Sun high all year, so it is warmer with a small seasonal swing; the higher-latitude place (Q, 50 degrees N) gets a low slanting winter Sun and a higher summer Sun, so it is cooler overall with a wider swing. Higher latitudes do not gain net warmth from long summer days, because those are paid back by long dark winters and weak slanting light. Matching coast and altitude controls those factors but not latitude, so the two are not identical. Equatorial places have the smallest, not the largest, seasonal swing.

11. (A) The Earth's axis keeps a fixed direction in space, so the tropics are not 'permanently tilted toward the Sun'; over the year each hemisphere tilts toward and then away. What keeps the tropics warm is that the Sun stays high overhead there every month, so rays land steeply in all seasons, which is Student 2's reasoning. Student 1 misdescribes the geometry. The two statements are not equivalent, since one names a false cause and the other the true cause. Nearness to the sea moderates ranges but does not explain why inland tropical places are also warm year round.

12. (D) Two effects weaken a slanting polar ray: the same bundle of light is smeared over a wider patch of ground, lowering energy per square metre, and a low-angle ray cuts through a longer slice of atmosphere, so more of its energy is scattered and absorbed before reaching the surface. Light does not change speed in a way that 'loses heat', and a slanting ray still reaches the ground. There is no inherently 'colder light' from the same Sun, and the ground does absorb sunlight. The Sun is not dimmer over the poles; it is the geometry and atmospheric path that matter.

13. (A) Air is warmed mainly from the ground up, and it thins with height; both effects make a high place cooler, so Town X at 2,000 m is colder than Town Y. The 'nearer the Sun' idea fails because 2,000 m is nothing against 150 million kilometres. Shared latitude controls one factor but not altitude, so the two towns are not equal. Climbing a mountain does not move a place toward the poles, so that reasoning is incoherent even though it reaches the right direction by accident.

14. (A) The base and summit share essentially the same latitude and the same horizontal distance from the sea, so neither can be the cause; what changes is height, and temperature falls steadily as you climb, letting jungle and snow coexist on one mountain. Latitude does not change meaningfully over a few horizontal kilometres. The summit's distance to the sea barely differs from the base's. The 'nearest the Sun is coldest' option is

self-contradictory, since being nearer the Sun could only warm a place; the summit is cold because of altitude.

15. (B) Applying 6.5 degrees per 1,000 m to 1,800 m gives a fall of about 11.7 degrees, so 32 minus roughly 12 leaves about 20 degrees, and the station is cooler because air is heated from the ground and thins with height. The 'warmer because nearer the Sun' option gets both the direction wrong and uses a negligible distance change. Altitude clearly affects temperature, so 'unchanged' is false. Climbing does not move a place toward the poles, and the correct lapse gives about 20, not 12, so that option errs in both cause and value.

16. (B) Water heats and cools slowly, so a coast (J) is moderated to a narrow range, while an interior (K) far from that buffer lets land heat fiercely in summer and shed heat fast in winter, widening the range at both ends; that is continentality. Latitude is stated equal, so it cannot be the hidden cause. Altitude is also stated equal, and interiors are not always higher than coasts. The sea moderates J by changing slowly, not by 'pumping' heat in and cold in opposite seasons, so that mechanism is misstated.

17. (D) Land responds quickly to gains and losses of heat, so an interior with no nearby sea to steady it warms intensely in summer and cools intensely in winter, driving both the high and the low further apart. The interior does not move toward or away from the Sun seasonally. A distant sea cannot reach inland to heat then chill a place; its moderating effect is felt only near the coast. A place's latitude is fixed and does not change between seasons.

18. (B) With latitude and altitude matched, a remaining climate factor is how well the moderating sea air actually reaches each place; a city open to onshore winds is steadied more than one shielded by terrain, so this could explain a small difference in range. Closeness to the Sun does not differ meaningfully between neighbouring coastal cities. Population is not the chapter's factor for the broad annual range. The Moon governs tides, not the seasonal temperature range, so it is irrelevant.

19. (C) The sea warms slowly, so Mumbai's coast is held cooler in summer, while inland Nagpur, lacking that buffer, heats rapidly and swelters; the pairing of cause to city is correct here. The sea moderates rather than reflecting extra heat onto the coast, so the 'fiercer Mumbai' option misstates the mechanism. Shared latitude controls only the latitude factor, leaving continentality free to differ, so the summers are not identical. Being inland does not change a city's latitude, so Nagpur is not 'nearer the Equator'.

20. (B) The sea changes temperature slowly, so a place near it is steadied and shows a small annual range; that nearness is the best explanation for a narrow gap. A dry continental interior is the opposite case, producing a large range. A high exposed plateau far from the coast also tends to swing widely. A high-latitude place has strongly changing sun-angles between summer and winter, which widens rather than narrows the range.

21. (A) The loo is hot and dry because it has crossed hot dry land, and the winter cold waves are cold because their air came over cold snowy high ground; both show that a wind delivers the character of the surface it has travelled across. Winds do not move a region to a new latitude, do not change the altitude of the ground beneath them, and cannot transport a place toward or away from the sea; the region stays put while the air, carrying its acquired qualities, moves over it.

22. (A) Air takes on the qualities of the surface it crosses, so a wind off a warm sea gains moisture while a wind off a desert stays dry; thus Wind A is the moister. Seas add moisture to passing air rather than stripping it, so 'drier' is backwards. A warm sea does not make a wind icy, and the reliable contrast asked for is moisture, not a guaranteed cold. The surface most certainly leaves its mark, which is the whole point of this chapter's wind discussion.

23. (D) One wind crossed cold snowy highlands and arrives chilly and dry; the other crossed a warm sea and arrives mild and damp; the contrast is set by the surfaces each travelled over. Latitude is not the active cause here, since the lesson is about what the winds picked up en route. 'Heavier and therefore colder' confuses a vague notion of weight with the real cause, the source region. The two are different air masses shaped by different paths, not one fickle wind 'changing its mind'.

24. (D) Flat openness lets a wind sweep across the plain unhindered, while a high ridge blocks or deflects it from the valley; whether a place is exposed or sheltered is set by the lie of the land, which is relief. Latitude is stated the same, so it cannot be the difference. The shelter comes from the ridge, not from differing nearness to the ocean. The Sun's position is identical for both at the same latitude, so it cannot explain why one is sheltered and the other exposed.

25. (C) Because a wind takes on the character of its path, the loo could only become cool and moist if it had crossed a cool sea that handed it moisture and lowered its heat. Crossing a bigger, hotter desert would make it even hotter and drier. Crossing hot summer plains would add heat, again the opposite. A surface wind cannot travel through empty space, and even if it could it would gain no moisture; the only path that produces cool and moist is over a cool sea.

26. (C) As air climbs it moves into lower pressure, expands, and cools; cooler air can hold less water vapour, so the surplus condenses into cloud and rain on the windward slope. Rising air expands and cools rather than being compressed and warmed, so the second option reverses the physics. Rising air does not 'dry out instantly without cooling'; the cooling is exactly what causes condensation. Air that has crossed the crest descends the far side, so it does not loop back to rain twice on the windward slope.

27. (B) The correct chain is: moist air is forced up the western slopes, drops its rain there, then spills over the crest as descending, drying air, so the Deccan sits in a rain shadow. The Deccan is not 'nearer the Equator', and heat does not 'repel' clouds. The coastal plain is

lower than the Ghats, not higher. Rivers do not manufacture rain; the dryness comes from the descending leeward air, not from a lack of rivers.

28. (A) Ladakh sits in the rain shadow of the Himalayas, which strip the winds of moisture, making it dry, and its great altitude makes it bitterly cold, so both halves of 'cold desert' are explained. There is no nearby sea pulling moisture away, and Ladakh is far from the North Pole. It is genuinely cold, not merely nicknamed so, unlike the hot Thar. Low latitude would tend to warm, not cool, and rivers evaporating does not cause the dryness; the rain shadow does.

29. (C) The winds reaching these hills are heavy with Bay of Bengal moisture, and the funnel-shaped terrain forces them sharply upward, cooling them so the vapour condenses into extraordinary rain. Dry Thar winds carry little moisture and cannot deliver record rain. Descending leeward air warms and dries, which suppresses rather than causes rain. The rain is a summer monsoon phenomenon, not cold winter Himalayan winds dropping snow.

30. (B) Each contrast pairs a place where moist air is forced up (wet) with a sheltered or lower place behind it (drier); relief lifting the monsoon winds explains the soaked windward slopes and the dry rain-shadow areas alike. Latitude cannot explain why two nearby places at the same latitude differ so sharply. Altitude alone fails, since the dry Deccan is not always lower than the wet coast. Distance from the sea alone fails, since the inland Deccan is drier, not wetter, than the coast.

31. (D) The Ghats cause the coast's heavy rain by forcing moist air up; remove them and the air is no longer lifted, so coastal rain falls and some moisture drifts inland to the formerly shadowed Deccan. The coast would rain less, not more, because the lifting that causes its rain is gone. Removing a mountain range does not freeze a tropical peninsula. Rainfall would clearly change, since the very mechanism creating the present pattern would be removed.

32. (A) Moist sea air rises on the windward slope, cools and rains, then crosses the crest and sinks dry and warming into the leeward valley, producing a wet side and a parched side within a few kilometres; this is the rain shadow. Latitude does not jump meaningfully over a few kilometres. The far valley is not closer to the Equator; the two sides share essentially the same latitude. Soil does not control rainfall; the difference is set by which side of the ridge the wind has crossed.

33. (A) The monsoon is driven by the difference in how fast land and sea heat and cool, which creates the seasonal high- and low-pressure zones the winds flow between, so removing it would kill the monsoon. The Moon's gravity governs tides, not the seasonal wind reversal. Rivers are far too small to generate a continent-sized pressure system. A fixed equatorial temperature is neither true nor the engine of the monsoon, which runs on the land-sea contrast.

34. (C) Intense summer heating makes the air rise over the land, creating low pressure, and air flows from the higher-pressure sea into that low, arriving from the south-west laden with moisture. Hot rising air gives low pressure, not high, so the two 'high pressure' options misname the system. Winds flow from high to low pressure, so they blow inland into the low, not outward to sea, which rules out the 'dry winds blow outward' option.

35. (B) The south-west monsoon picks up moisture over warm seas before reaching land, so it rains heavily, whereas the winter north-east monsoon starts over dry land and heads seaward, so it is mostly dry. The first wrong option reverses the geography, since the south-west winds come off the sea, not the Thar. Cold winds do not 'always carry more rain'; moisture, not coldness, brings rain. Monsoon winds blow horizontally from sea to land, not vertically down from the upper atmosphere.

36. (A) Although the winter monsoon generally blows off dry land, on its way to the Tamil Nadu coast it crosses the Bay of Bengal, gaining the moisture it then drops as rain. The Thar is a dry desert and supplies no rain-making moisture. Winds blowing down the Himalayas arrive cold and dry. The Arabian Sea is on the west; the moisture for this south-eastern coastal rain comes from the Bay of Bengal to the east.

37. (B) In summer the hot land is low pressure and draws in sea winds; in winter the cold land becomes high pressure and pushes winds out to sea; this seasonal flip of the land-sea pressure pattern reverses the winds. The Earth's axial tilt keeps a fixed direction in space and does not flip twice a year. The Western Ghats are stationary and cannot swing with the seasons. The Moon's orbit does not switch direction and does not drive the monsoon.

38. (D) The year runs from the dry winter into the building hot dry summer, then the rains burst as the south-west monsoon, and finally the monsoon retreats before winter returns; this is the correct order. The rains do not precede the summer heat, so 'rainy then hot' is out of sequence. The retreating phase comes after, not before, the rains, ruling out the option that places the retreat first and the option that places the retreat before the rains.

39. (B) A daytime sea breeze arises because sun-warmed land forms low pressure and draws cooler sea air ashore; the summer monsoon does the same on a continental scale, which is why the analogy works. The Moon drives tides, not the pressure-driven inflow of sea air. Rivers are far too small to set up a country-sized wind system. Snowmelt does not power the monsoon, which is driven by the land-sea heating contrast just like a sea breeze.

40. (A) The south-west monsoon arrives first at the southern tip in early June and then spreads northward, blanketing the country by about mid-July. It does not begin in the north and creep south to reach Kerala last. It moves inland from the seas over the land, not from the coast out to sea. And it does not arrive everywhere at once; its northward advance takes several weeks.

41. (C) After a long, hot, dry build-up, the moisture-laden south-west winds sweep in and pour down heavy rain, and the sharp contrast with the preceding dryness makes the onset feel like a sudden 'burst'. The Sun does not switch off. Rivers do not lift their water into the sky to make rain; rain comes from the moist incoming winds. The burst is the arrival of the wet summer monsoon, not the dry winter north-east monsoon.

42. (B) The retreating monsoon is the transition when the south-west monsoon loses strength and pulls back, linking the rainy season to the coming dry winter. The first bursting over Kerala is the onset, the opposite end of the season. The pre-rain build-up of heat is the hot dry summer, which comes before the rains. Rivers do not climb back into the mountains; the term refers to the withdrawing winds.

43. (C) The monsoon is so powerful in shaping farming and daily life in India that the rains are recognised as a season in their own right, giving the idea of a 'fifth season'. The five-season idea is not a European import; it grows from India's monsoon-ruled year. India does have a winter, so the 'no winter' claim is false. Festivals follow the seasons rather than each demanding a separate one, so they do not create the fifth season.

44. (C) The six ritus split the year into finer slices that match India's monsoon-shaped rhythm, capturing transitions the four-season model blurs. They are an ancient classification, not a recent tourist invention. Far from ignoring the monsoon, they are organised around the rains and the seasons it produces. They are not the four European seasons renamed; there are six, attuned to local conditions.

45. (A) These harvest and seasonal festivals fall when the climate-driven farming cycle reaches key moments, showing that climate shapes agriculture and so the rhythm of cultural life. Festivals are responses to the seasons; they do not command the rains to arrive. Climate clearly does influence culture here, so 'no effect' is contradicted by the very examples given. The festivals follow the farming and monsoon calendar, not latitude alone.

46. (B) Because the tilted Earth orbits the Sun, each part of the year brings the same angle of sunlight as the year before, so the seasons recur annually. The Sun does not actually heat up in summer; the change is in the angle of its rays on the tilted Earth. The monsoon reversal is a consequence of seasonal heating, not a choice keyed to festivals. River floods follow the seasons; they do not cause them.

47. (B) Srinagar is cold because two factors reinforce each other: a high latitude gives it a low, slanting Sun, and its high mountain altitude lowers temperatures further. It is at a high, not low, latitude, and it is far from the moderating sea, so 'low latitude and nearness to the sea' is doubly wrong. It is nowhere near the Equator, so 'overhead Sun' is false. Warm moist sea winds do not steadily blow over an inland mountain valley.

48. (A) Ladakh's cold dryness comes from great altitude and the rain shadow of the Himalayas; Mumbai's small range comes from the moderating sea; Cherrapunji's deluge

comes from relief forcing moist winds up; Kanniyakumari's steady warmth comes from its low, near-equatorial latitude. The other matchings misassign factors: calling Ladakh 'low latitude' or 'near the sea' is wrong, pinning Mumbai's mildness on altitude or latitude ignores the sea, and crediting Cherrapunji to distance from the sea or altitude alone misses the lifting of moist winds.

49. (B) When the monsoon fails, fields go short of water, crops suffer, harvests shrink and food prices climb; this is the realistic chain of effects. A failed monsoon means too little rain, so rivers do not overflow and harvests do not boom. Weak rains do not warm winters or grow glaciers; they cause scarcity, not abundance. The ritus are a calendar of seasons; a poor rain year does not erase them.

50. (D) So much Indian agriculture relies on the monsoon that a weak season cuts harvests, lowers the incomes of farming families, and pushes up food prices, striking the whole economy. Festivals are not legally cancelled in dry years. The sea's moderating effect on coastal cities does not switch off when the rains are poor. The Himalayas do not block trade because of a dry year; the real link runs through farm output.

51. (C) When warm Pacific water shifts and rearranges tropical winds (an El Nino pattern), the Indian monsoon tends to weaken, so India gets less rain and crops and water are stressed. A weakened monsoon means drought-prone conditions, not record floods. Ocean warming in the Pacific does not raise the height of the Himalayas. The coast's moderation by the sea is unrelated to a Pacific warming event, so that link does not follow.

52. (A) The periodic warming of the central and eastern Pacific that disturbs tropical winds and can weaken India's monsoon is called El Nino. The loo is a local hot summer wind of north-west India, not a Pacific event. A rain shadow is the dry leeward side of a mountain, unrelated to the Pacific. The burst is the dramatic onset of the monsoon rains, not the Pacific warming that disrupts them.

53. (B) A cyclone is a deep low-pressure system; warm moist air rises at the centre, pressure falls there, and surrounding air with its moisture rushes in and spirals upward, feeding the storm. High central pressure would push air outward and could not draw the storm together. Equal pressure everywhere would generate no inflow and so no cyclone. Being merely 'colder than land in winter' does not describe the pressure condition that creates a cyclone.

54. (D) The sudden calm and clear sky mark the cyclone's eye passing overhead; the violent eyewall winds will return, often from the opposite direction, so one must stay sheltered. The storm has not ended; the eye is only its centre. It has not become a harmless breeze; the lull is temporary. The eye itself is the calmest, not the most destructive, part, so fleeing through it on that belief is mistaken; the danger is the returning eyewall.

55. (B) The fiercest winds spin in the eyewall, the ring of cloud wrapped around the calm centre, so that is where danger is greatest. The eye itself is calm, not strong. Far beyond the storm the winds are weak. After the storm has fully passed the air is calm, so the worst winds are not 'only after' it ends; they are in the eyewall during the storm.

56. (D) The eye's sudden calm and clear sky fool people into thinking the storm is over, but the eyewall's violent winds resume as soon as the eye passes, which is why going outside is dangerous. The eye is genuinely calm, not secretly the strongest part. A cyclone can still be intense after the eye passes, so 'can never strengthen again' is false. The eye does not mark the cyclone leaving the sea; it is simply its calm centre.

57. (C) The natural greenhouse effect retains some of the Sun's heat, keeping Earth warm enough to support life; without it the planet would be far too cold. The gases do not block all sunlight; they let sunlight in and trap part of the outgoing heat. They warm rather than cool the planet, so 'below freezing' is the opposite of their effect. They do not directly create the monsoon, which is driven by the land-sea heating contrast.

58. (D) Burning coal releases carbon dioxide, which traps extra heat, shifting the climate, disrupting the monsoon and causing the crop to fail; this is the correct cause chain. Burning coal warms rather than cools, so 'sea cools, monsoon strengthens, crop thrives' is wrong at the first link. Burning coal does not freeze rivers. Burning coal does not grow forests; the chain ends in a failed crop from a disrupted monsoon, not a flood from new forests.

59. (A) Mitigation means reducing the cause of warming, so switching from coal to solar and wind, which cuts emissions, is a clear mitigation action. Building sea walls, stockpiling supplies for drought, and relocating homes from a flood zone all help communities cope with or withstand impacts, which makes them resilience measures, since none of them reduces greenhouse-gas emissions.

60. (C) Since the 1800s, climate change has been driven mainly by human activities, above all the burning of fossil fuels that release heat-trapping gases. Natural processes alone do not account for the recent rapid warming. The Earth is not drifting measurably farther from the Sun. Volcanic eruptions can affect climate briefly but are not the main long-term driver, which is human-caused emissions.